Supersymmetric and Shape-Invariant Generalization for Non-resonant Jaynes-Cummings Systems

A. N. F. Aleixo *
Department of Physics, University of Wisconsin
Madison, Wisconsin 53706 USA †

A. B. Balantekin ‡,
Max-Planck-Institut für Kernphysik, Postfach 103980, D-69029 Heidelberg, Germany §

M. A. Cândido Ribeiro ¶,
Departamento de Física - Instituto de Biociências, Letras e Ciências Exatas
UNESP, São José do Rio Preto, SP - Brazil

(November 13, 2018)

Abstract

A class of shape-invariant bound-state problems which represent transitions in a two-level system introduced earlier are generalized to include arbitrary energy splittings between the two levels. We show that the coupled-channel Hamiltonians obtained correspond to the generalization of the non-resonant Jaynes-Cummings Hamiltonian, widely used in quantized theories of laser. In this general context, we determine the eigenstates, eigenvalues, the time evolution matrix, and the population inversion matrix factor.

Typeset using LaTeX

*Electronic address: aleixo@nucth.physics.wisc.edu
†Permanent address: Instituto de F´ısica, Universidade Federal do Rio de Janeiro, RJ - Brazil.
‡Electronic address: baha@nucth.physics.wisc.edu
§Permanent address:Department of Physics, University of Wisconsin, Madison, Wisconsin 53706 USA
¶Electronic address: macr@df.ibilce.unesp.br

I INTRODUCTION

The integrability condition called shape-invariance originates in supersymmetric quantum mechanics [1,2]. The separable positive-definite Hamiltonian $\hat{H}_1 = \hat{A}^\dagger \hat{A}$ is called shape-invariant if the condition

$$\hat{A}(a_1)\hat{A}^\dagger(a_1) = \hat{A}^\dagger(a_2)\hat{A}(a_2) + R(a_2) \tag{1}$$

is satisfied [3]. In this equation a_1 and a_2 represent parameters of the Hamiltonian. The parameter a_2 is a function of a_1 and the remainder $R(a_1)$ is independent of the dynamical variables such as position and momentum. Even though not all exactly-solvable problems are shape-invariant [4], shape invariance, especially in its algebraic formulation [5-7], has prove, to be a powerful technique to study exactly-solvable systems.

II THE GENERALIZED NON-RESONANT JAYNES-CUMMINGS HAMILTONIAN

II.1 The Resonant Limit

II.2 The Standard Jaynes-Cummings Limit

III THE TIME EVOLUTION OF THE SYSTEM

To study the time-dependent Schrödinger equation for a Jaynes-Cummings system in non-resonant situation

$$i\hbar\frac{\partial}{\partial t}|\Psi(t)\rangle = (\hat{H}_0 + \hat{H}_{int})|\Psi(t)\rangle \tag{2}$$

we can write the wavefunction as

$$|\Psi(t)\rangle = exp(-i\hat{H}_0 t/\hbar)|\Psi_i(t)\rangle, \tag{3}$$

and, by substituting this into Schrödinger equation and taking into account the commutation property between $\hat{H}_0$ et $\hat{H_{int}}$, we obtain

$$i\hbar\frac{\partial}{\partial t}|\Psi_i(t)\rangle = \hat{H}_{int}|\Psi_i(t)\rangle \tag{4}$$

We introduce the evolution matrix $\hat{U}_i(t,0)$:

$$|\Psi_i(t)\rangle = \hat{U}_i(t,0)|\Psi_i(0)\rangle \tag{5}$$

which satisfies the equation

$$i\hbar\frac{\partial}{\partial t}\hat{U}_i(t,0) = \hat{H}_{int}\hat{U}_i(t,0), \tag{6}$$

that is, in matrix form, written as

$$i\hbar\begin{bmatrix} \hat{U}'_{11} & \hat{U}'_{12} \\ \hat{U}'_{21} & \hat{U}'_{22} \end{bmatrix} = \alpha\begin{bmatrix} \beta & \hat{T}\hat{B}_- \\ \hat{B}_+\hat{T}^\dagger & -\beta \end{bmatrix}\begin{bmatrix} \hat{U}_{11} & \hat{U}_{12} \\ \hat{U}_{21} & \hat{U}_{22} \end{bmatrix} \tag{7}$$

where the primes denote the time derivative. One fast way to diagonalize the evolution matrix differential equation is by differentiating Eq. (3.5) with respect to time. We find

$$i\hbar\frac{\partial^2}{\partial t^2}\hat{U}_i(t,0) = \hat{H}_{int}\frac{\partial}{\partial t}\hat{U}_i(t,0) = \frac{1}{i\hbar}\hat{H}^2_{int}(t,0) \tag{8}$$

which can be written as

$$\begin{bmatrix} \hat{U''}_{11} & \hat{U''}_{12} \\ \hat{U''}_{21} & \hat{U''}_{22} \end{bmatrix} = -\begin{bmatrix} \omega_1 & 0 \\ 0 & \omega_2 \end{bmatrix}\begin{bmatrix} \hat{U}_{11} & \hat{U}_{12} \\ \hat{U}_{21} & \hat{U}_{22} \end{bmatrix} \tag{9}$$

where

$$\hbar\hat{\omega}_1 = \alpha\sqrt{\hat{T}\hat{B}_-\hat{B}_+ + \beta^2} = \sqrt{\hbar\Omega\hat{H}_2 + (\hbar\Delta)^2} \tag{10}$$

$$\hbar\hat{\omega}_2 = \alpha\sqrt{\hat{B}_-\hat{B}_+ + \beta^2} = \sqrt{\hbar\Omega\hat{H}_1 + (\hbar\Delta)^2} \tag{11}$$

Since by initial conditions $\hat{U}_i(0,0) = \hat{I}$, , then we can write the solution of the evolution matrix differential equation (3.7) as

$$\hat{U}_i(t,0) = \begin{bmatrix} \cos(\omega_1 t) & \sin(\omega_1 t)\hat{C} \\ \sin(\omega_2 t)\hat{D} & \cos(\omega_2 t) \end{bmatrix} \tag{12}$$

and the $\hat{C}$ the $\hat{D}$ operators can be determined by the unitarity conditions

$$\hat{U}^\dagger_i(t,0)\hat{U}_i(t,0) = \hat{U}_i(t,0)\hat{U}^\dagger_i(t,0) = \hat{I}. \tag{13}$$